

Skills Summary

Quadrant and Parameter-Orientation Errors

Use this reference while completing your worksheet or when studying.

The one habit that prevents all of these: a calculator returns a *number*; you supply the *position*. Before writing an angle, decide the quadrant from the signs of x and y (or from the sign of r); before writing a parameter answer, decide how far around the curve the parameter runs.

1. Rectangular to polar: the quadrant decision

Rule: $r = \sqrt{x^2 + y^2}$, reference angle $\alpha = \arctan \left| \frac{y}{x} \right|$, then place α :

QI: $\theta = \alpha$ QII: $\theta = 180^\circ - \alpha$ QIII: $\theta = 180^\circ + \alpha$ QIV: $\theta = 360^\circ - \alpha$

$\arctan$ only ever returns $(-90^\circ, 90^\circ)$ — QI or QIV — so it can never place a QII or QIII point by itself.

Example: Write $(-4, 3)$ in polar form with $0^\circ \leq \theta < 360^\circ$.

Step 1. $x < 0$, $y > 0$, so the point is in Quadrant II — take the reference angle and reflect it, do not trust the raw $\arctan$.

Step 2. $\alpha = \arctan \left(\frac{3}{4} \right) = 36.87^\circ$, so $\theta = 180^\circ - 36.87^\circ$; also $r = \sqrt{16 + 9} = 5$. $\Rightarrow \theta = 143.13^\circ$

Try it: Write $(5, -12)$ in polar form with $0^\circ \leq \theta < 360^\circ$. (Which quadrant?)

check: $\theta = 110.93^\circ$, $r = 13$

2. Negative r and equivalent names

Rule: $(-r, \theta)$ is the same point as $(r, \theta + 180^\circ)$ — go out along θ , then straight through the pole.

Conversion formulas take the *signed* r : $x = r \cos \theta$, $y = r \sin \theta$. To rename a point with a positive r , add or subtract 180° — whichever keeps the angle inside $[0^\circ, 360^\circ)$.

Example: Find the x -coordinate of the polar point $(-3, 40^\circ)$.

Step 1. The sign of r is part of the data, so substitute $r = -3$ rather than dropping the minus sign.

Step 2. $x = r \cos \theta = -3 \cos 40^\circ = -3(0.766\dots)$ (equivalently, the point is $(3, 220^\circ)$). $\Rightarrow x = -2.30$

Try it: Find the y -coordinate of the polar point $(-5, 110^\circ)$.

check: $y = -4.70$

3. Parametric orientation: which way, from where

Rule: the same curve has many parametrizations; the parameter carries the *starting point* and the *direction*, which the Cartesian equation does not.

To read orientation, tabulate $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ and watch the points move. Swapping sin and cos (or replacing t by $-t$) keeps the curve and reverses the trip.

Example: For $x = 4 \sin t$, $y = 4 \cos t$, find x at $t = \frac{\pi}{6}$.

Step 1. Substitute into the rule for x as written — the sin belongs to x here, which is exactly the swap that trips people up.

Step 2. $x = 4 \sin \frac{\pi}{6} = 4 \left(\frac{1}{2}\right)$. (At $t = 0$ this curve is at $(0, 4)$ and moves right: clockwise.)
 $\Rightarrow x = 2$

Try it: For $x = 4 \sin t$, $y = 4 \cos t$, find x at $t = \frac{\pi}{3}$.

check: $x = 2$

4. Every parameter value, not just the first

Rule: arcsin, arccos and arctan each return ONE value. On a full trip round a closed curve, a height or an x -value is usually reached **twice**.

$\sin t = k$: $t = \alpha$ and $180^\circ - \alpha$. $\cos t = k$: $t = \alpha$ and $360^\circ - \alpha$.

Then keep only the values inside the stated parameter interval.

Example: On $0^\circ \leq t < 360^\circ$, solve $2 \cos t - 1 = 0$.

Step 1. Cosine is positive in Quadrants I and IV, so expect two parameter values, not one.

Step 2. $\cos t = \frac{1}{2}$ gives the reference angle 60° ; the QIV partner is $360^\circ - 60^\circ$.

$\Rightarrow t = 60^\circ, 300^\circ$

Try it: On $0^\circ \leq t < 360^\circ$, solve $2 \sin t + 1 = 0$.

check: $t = 210^\circ, 330^\circ$